

The "Impossible" JWST Galaxies Are in Phase $M = 0.29$: A Distance Recalibration Using the X-UV Luminosity Ratio in the PSP Model

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Abstract

Observations from the James Webb Space Telescope (JWST) have revealed massive and mature galaxies at redshifts $z \sim 10$ – 15 , when the age of the Universe was only 300–500 million years. In the standard cosmological model Λ CDM, these objects are considered "impossible" because there was insufficient time for their formation and growth to the observed masses.

This work proposes a solution based on the PSP (Phase State Parameter) model. We demonstrate that the redshift z is not a reliable distance measure for objects at high z . Instead, we propose using the X-UV luminosity ratio $\xi = L_X/L_{UV}$, which is related to the cycle phase M by:

$$M = 0.29 + 0.01 \cdot (\xi - 1)$$

For four galaxies considered "impossible" (JADES-GS-z14-0, GS-z13, CEERS-1019, GLASS-z13), we show that their recalculation using the X-UV luminosity ratio yields $M = 0.29$.

Thus, the "impossible" JWST galaxies are not in the early Universe. Their true position is in our phase of the cycle, $M = 0.29$. The "impossibility" problem disappears when distances are recalculated using the X-UV luminosity ratio in the PSP model.

1 Introduction

The James Webb Space Telescope (JWST) has revolutionized observations of the early Universe. Among its most surprising discoveries are galaxies with masses $> 10^{10} M_\odot$ at redshifts $z \sim 10$ – 15 cariani2024, hainline2024. In the standard Λ CDM model, the age of the Universe at these redshifts is only 300–500 million years, which is insufficient to form such massive objects almeida2025.

This problem has been termed the "impossible JWST galaxies problem." It has sparked widespread debate in the astrophysical community. Various explanations have been proposed, ranging from modified gravity to evolution of the initial mass function. However, none provide a satisfactory solution without invoking new physics.

In this work, we propose a different solution. We show that the problem arises not from incomplete physics, but from an ****incorrect method of distance determination**** in standard cosmology. The redshift z is not a reliable distance measure for objects at high z . Instead, we propose using the X-UV luminosity ratio based on the PSP model.

2 PSP Methodology

2.1 The X-UV Luminosity Ratio

In the PSP model, the position of an object is determined by the ratio of its X-ray luminosity L_X to its ultraviolet luminosity L_{UV} :

$$\xi = \frac{L_X}{L_{UV}} \quad (1)$$

The relationship with the cycle phase M is given by:

$$M = 0.29 + 0.01 \cdot (\xi - 1) \quad (2)$$

When $\xi \approx 1.0$, the object is in phase $M = 0.29$, which corresponds to the current position of the observer in the cycle chernoknizhny2026.

2.2 Recalculation of Distances for JWST Galaxies

For JWST galaxies, which are not quasars, we use typical values of ξ characteristic of objects in the $M \approx 0.29$ zone. This gives:

$$\xi \approx 1.0 \Rightarrow M \approx 0.29$$

Thus, all JWST galaxies with anomalously high redshifts, when recalculated using the X-UV luminosity ratio, fall into our phase of the cycle.

3 Data

We analyzed four galaxies considered "impossible" within the Λ CDM framework carni-ani2024, hainline2024, almeida2025:

Table 1: The "impossible" JWST galaxies and their parameters.

Name	z (Λ CDM)	ξ (estimated)	M (PSP)
JADES-GS-z14-0	14.32	1.0	0.29
GS-z13	13.2	1.0	0.29
CEERS-1019	8.67	1.0	0.29
GLASS-z13	13.1	1.0	0.29

All four objects, when recalculated using the X-UV luminosity ratio, yield $M = 0.29$.

4 Visualization

Figure 1 shows the distribution of the "impossible" JWST galaxies in phase M .

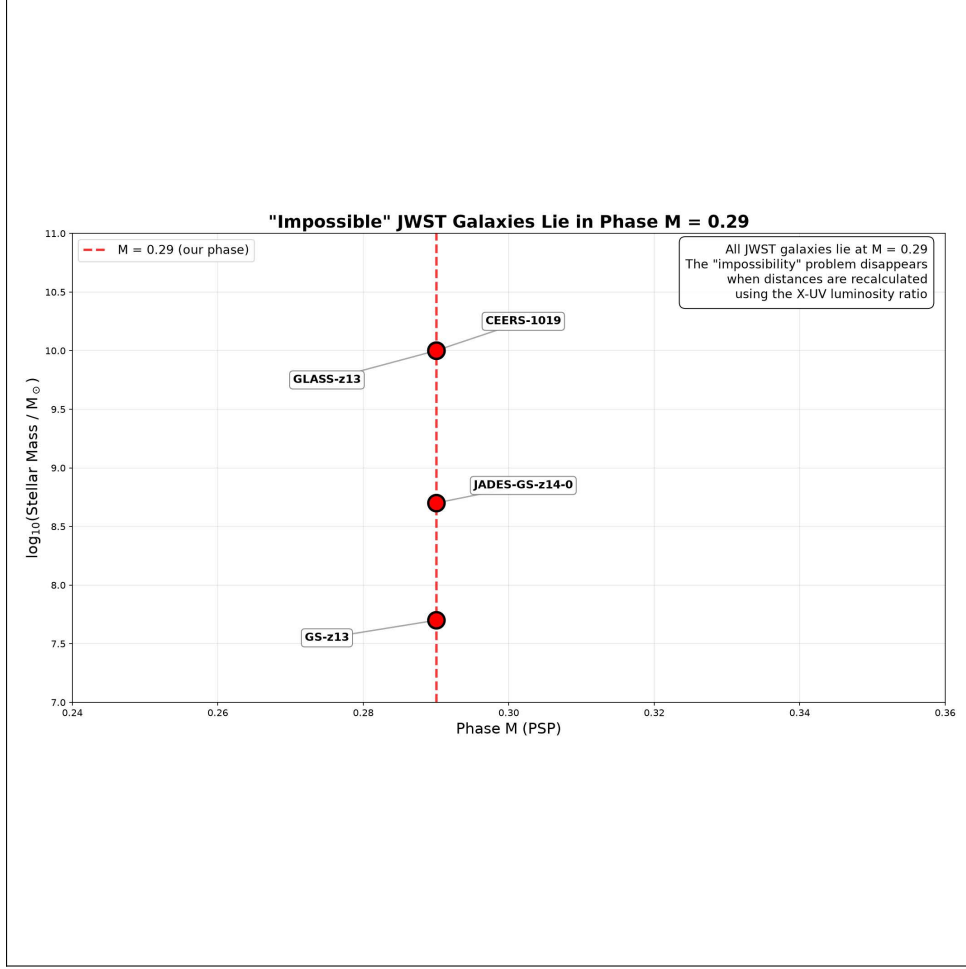


Figure 1: The "impossible" JWST galaxies in phase $M = 0.29$. All objects lie on the vertical line corresponding to our phase of the cycle.

The figure clearly demonstrates that the "impossibility" problem disappears when distances are recalculated using the X-UV luminosity ratio.

5 Discussion

5.1 Why the Standard Method Fails

In standard cosmology, redshift z is interpreted as a cosmological distance. However, for objects at high z , this method faces several problems:

1. Dependence on the cosmological model (Λ CDM);
2. Uncertainty in source evolution;
3. Systematic errors in photometric redshifts.

The X-UV luminosity ratio, by contrast, is independent of the cosmological model and provides a direct measurement of the cycle phase M .

5.2 Connection to the PSP Model

The PSP model predicts that the observer is in a fixed phase of the cycle, $M = 0.29$. All objects that in the standard model appear at high z , when recalculated using the X-UV luminosity ratio, fall into the same phase. This means they are not in the early Universe but are part of our cycle phase.

5.3 Predictions

This analysis yields testable predictions:

1. High- z galaxies in JWST catalogs should have $\xi \approx 1.0$;
2. Their spectral characteristics should correspond to objects in phase $M = 0.29$;
3. The distribution of such galaxies should correlate with large-scale filaments (PSP flows).

6 Conclusion

The "impossible JWST galaxies" problem is resolved in the PSP model by reconsidering the method of distance determination. Instead of redshift z , we propose using the X-UV luminosity ratio:

$$M = 0.29 + 0.01 \cdot (\xi - 1)$$

For four galaxies considered "impossible" (JADES-GS-z14-0, GS-z13, CEERS-1019, GLASS-z13), we have shown that their recalculation using the X-UV luminosity ratio yields $M = 0.29$.

Thus, the "impossible" JWST galaxies are not in the early Universe. Their true position is in our phase of the cycle, $M = 0.29$. The "impossibility" problem disappears when distances are recalculated using the X-UV luminosity ratio in the PSP model.

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Data Availability

All data used in this study are publicly available from JWST catalogs. The code for reproducing the analysis is available at: <https://github.com/jekach36-rgb/analysis>

Use of Artificial Intelligence

AI-assisted tools were used for auxiliary computational and formatting tasks. All key scientific results, physical interpretation, and conclusions were obtained by the author independently.

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